Data Communication and Computer Networks

Chapter 5 | Application Layer

Unit 5 - Client-server model, DNS, e-mail, FTP, remote login, HTTP and WWW (6 hrs)

Syllabus covered

Client-server model and the socket interface; Domain Name System; electronic mail - SMTP, POP, IMAP; file transfer - FTP; remote logging - Telnet; HTTP and the World Wide Web.

What is inside

Section	Question type	Marks	Count
Section A	Short answer - definitions, one-liners, small numericals	2	27
Section B	Long answer - full explanations with tables and diagrams	7	8

How to use this bank

- This chapter is memory work: ports, protocols and their transport. Learn the port table first - it answers many 2-mark questions on its own.
- Every 7-mark answer here is a diagram plus a walk-through of the same four steps: request, resolve or connect, transfer, close.
- Keep DNS, SMTP, FTP and HTTP separate in your notes; examiners penalise mixed answers.

Source tags

- **PYQ <year>** - asked in a C. V. Raman Global University end-semester paper (2022, 2023, 2024-25 or the 2024-25 backlog paper).
- **Model** - from the printed unit-wise model question paper.
- **Assignment** - from the course assignment for this unit.
- **Notes** - from the lecture slides, not yet asked but examinable.

Section A | 2-mark questions

Answer in two to four lines. Each point below is one mark-bearing idea, so a single line written in the exam earns the mark.

A1. What is the client-server model? [PYQ 2022]

- A **client** is a program running on the local machine that requests a service; a **server** is a program on a remote machine that provides it.
- One server can serve many clients at the same time - web, mail and file servers are the standard examples.
- Almost every application-layer protocol in this unit - DNS, HTTP, FTP, SMTP - follows this model.

A2. State the advantages and disadvantages of the client-server model. [Notes]

- **Centralised**: backup, administration and shared resources are in one place.
- **Secure**: access to resources is controlled centrally.
- **Performance and scalability**: a dedicated server improves overall performance, and new nodes can be added at any time.
- **Disadvantages**: **traffic congestion** at the server, a **single point of failure** when the server is down, and data packets may be modified in transit.

A3. What is a socket? [Notes]

- A socket is the **software interface** between the application layer and the transport layer inside one host.
- A process sends messages into, and receives them from, the network through this interface.
- A socket is identified by the pair (IP address, port number), so a connection is identified by four values: source IP and port, destination IP and port.

A4. Define DNS. [PYQ 2022; PYQ 2023]

- The Domain Name System is the hierarchical naming system that translates **human-readable domain names into IP addresses**, and back.
- It works on the **client-server model**, uses the **UDP** protocol on **port 53**, and has a hierarchical domain-based naming scheme.
- It is needed because users remember names, while routers need numbers.

A5. List the different types of name servers used in DNS. [PYQ 2022; PYQ 2023; PYQ 2024-25]

- **Root name server** - at the top of the tree; it knows the servers of the top-level domains.
- **Primary (authoritative) name server** - holds the zone file for a domain and is the original source of its records.
- **Secondary name server** - keeps a copy of the zone file from the primary, for redundancy and load sharing.
- **Local (caching) name server** - the one each ISP or organisation runs; it answers from its cache and queries others when it must.

A6. Explain the working of DNS briefly. [PYQ 2023]

1. The user program sends a name query to a library routine called the **resolver**.
2. The resolver checks its **local cache**; if the name is found, the IP address is returned at once.
3. Otherwise it queries the **local DNS server**, which checks its own cache and zone file.
4. If the local server cannot answer, it queries a **higher-level server** - root, then the top-level domain server, then the authoritative server - and passes the answer back.
5. The resolver caches the answer for its TTL and returns the result to the application.

A7. Write two examples each for a generic domain and a country domain in DNS. [PYQ 2024-25]

- **Generic domains** - three-character labels describing the type of organisation: **.com** (commercial), **.org** (non-profit), **.edu** (education), **.gov** (government), **.net** (network provider), **.mil** (military), **.int** (international).
- **Country domains** - two-character labels: **.in** (India), **.us** (United States), **.uk** (United Kingdom), **.jp** (Japan).
- Example: www.cgu-odisha.ac.in uses the country domain .in under .ac.

A8. What is the inverse domain? [Notes]

- The inverse domain maps an **IP address back to a domain name** - a reverse lookup.
- It is used when a server needs to know the name of the client that sent a request, for logging or access control.
- For IPv4 the zone is **in-addr.arpa**, with the address written in reverse order.

A9. What is the domain name space? [Notes]

- The domain name space is the whole naming tree, with the root at the top and labels separated by dots.
- It is divided into three sections: **generic domains**, **country domains** and the **inverse domain**.
- Each node of the tree is an index into the DNS database, and its label must be unique within its parent.

A10. Define SMTP and state its characteristics. [PYQ 2022]

- Simple Mail Transfer Protocol sets the rules for **sending e-mail** between servers; it is the push protocol of e-mail.
- It is an **application-layer** protocol that uses **TCP port 25**, so it is connection-oriented and reliable.
- It is a **pure text protocol**: it can send text only, which is why MIME was added for images, audio and video.
- Mail is read with a separate access protocol, POP3 or IMAP.

A11. Name two mail access protocols and explain them. [PYQ 2024-25]

- **POP3** (Post Office Protocol, version 3) - the client connects to the mail server on **TCP port 110**, sends a user name and password, lists the messages and downloads them one by one; the mail normally leaves the server, so it is designed for one device.
- **IMAP** (Internet Message Access Protocol) - uses **port 143**; mail stays on the server and is synchronised across devices, so reading, deleting and folders behave the same everywhere.

A12. Differentiate POP3 and IMAP. [Notes]

Point	POP3	IMAP
Port	110 (995 secure)	143 (993 secure)
Where mail is kept	Downloaded to the device	Stays on the server
Synchronisation	Not synchronised	Synchronised across devices
Best for	One device, offline reading	Many devices, webmail
Server storage	Low	High

A13. What is MIME and why is it needed? [Notes]

- Multipurpose Internet Mail Extensions extends SMTP, which can carry text only.
- It defines how **non-text content** - images, audio, video and attachments - is encoded into ASCII so that it can pass through SMTP.
- The receiver's user agent decodes the MIME part back into the original file.

A14. State the functionality of FTP and the two connections it uses. [PYQ 2022; PYQ 2024-25]

- FTP transfers files from one host to another and hides the differences between their file systems, text conventions and directory structures.
- **Control connection** - on **TCP port 21**, kept open for the whole session, carrying commands and replies.
- **Data connection** - on **TCP port 20**, opened for each file transfer and closed when the file is done.
- Splitting control from data is what allows commands such as abort while a transfer is in progress.

A15. State the advantages and disadvantages of FTP. [Notes]

- **Advantages**: fast, efficient (a transfer can be resumed or aborted without redoing everything), and secure because the server requires a user name and password.
- **Disadvantages**: a file size limit of about 2 GB, no simultaneous transfer to multiple receivers, and it is not compatible with every system.

A16. Define a proxy server and explain its relation to HTTP. [PYQ 2022; PYQ 2023]

- A proxy server is an **intermediary** between the client and the origin server: the client's HTTP request goes to the proxy, which fetches the object and returns it.
- It **caches** responses, so repeated requests are served locally - this cuts delay and saves bandwidth.
- It also provides **access control, filtering and anonymity**, since the origin server sees the proxy's address, not the client's.
- HTTP is designed for it: a proxy is simply an HTTP server that also acts as a client.

A17. Discuss the relationship of HTTP with the WWW. [PYQ 2024-25]

- The **WWW** is the collection of linked hypertext documents; **HTTP** is the protocol that transfers them.
- A browser is an HTTP client: it sends a request for a URL and the server returns the object, which the browser interprets and displays.
- HTTP can carry plain text, hypertext, audio and video, which is why the web is multimedia.
- In short: WWW is the *content*, HTTP is the *delivery*, and the URL is the *address*.

A18. Name the two types of HTTP messages and their parts. [Notes]

- **Request message** - sent by the client: request line (method, URL, version), headers, and sometimes a body.
- **Response message** - sent by the server: status line (version, status code, phrase), headers, and sometimes a body.
- Common methods: GET, POST, PUT, DELETE; common codes: 200 OK, 404 Not Found, 500 Server Error.

A19. Differentiate non-persistent and persistent HTTP connections. [Notes]

Point	Non-persistent	Persistent
Connection	Closed after each object	Kept open for many objects
TCP setup cost	Paid for every object	Paid once
Delay	Higher for a page with many objects	Lower
Version	HTTP/1.0 by default	HTTP/1.1 default

A20. What is a URL? Name its four parts with an example. [Notes]

- A Uniform Resource Locator is the standard way of specifying information on the Internet.
- Four parts: **method** (the protocol), **host** (the computer), **port** (optional), and **path** (the file).
- Example https://www.cgu-odisha.ac.in/ - method https, host www.cgu-odisha.ac.in, port 443 by default, path /.

A21. What is the World Wide Web? [PYQ 2022]

- The WWW is a method of browsing all the online data present on the Internet - a universe of documents on different websites, reachable from anywhere.
- It is built on the **client-server model** and **hypertext**: the client is the browser, the server is the web server, and links join the documents.
- Its original form was defined by HTTP 0.9 and HTML 1.0; the modern model is a generic resource exchange in which the client can also post data to the server.

A22. What is remote logging? Explain NVT and the pseudoterminal driver. [Notes]

- Logging into a local time-sharing system is **local login**; accessing an application on a remote machine is **remote login**, done by Telnet client and server programs.
- The client converts keystrokes into **Network Virtual Terminal** characters - a universal character set - so that any terminal can talk to any host.
- At the remote end the Telnet server converts NVT back into the remote system's characters.
- A **pseudoterminal driver** is needed because the remote operating system expects input from a terminal driver, not from a server; the driver pretends the characters came from a terminal.

A23. Mention any four application-layer protocols used in data communication. [PYQ 2023]

- **HTTP** - transfers web documents, TCP port 80.
- **SMTP** - sends e-mail, TCP port 25.
- **FTP** - transfers files, TCP ports 20 and 21.
- **DNS** - resolves names to addresses, UDP port 53.
- Also common: Telnet (23), POP3 (110), IMAP (143), HTTPS (443), SNMP.

A24. Give the port numbers of the common application protocols. [Notes]

Protocol	Port	Transport
FTP data / control	20 / 21	TCP
Telnet	23	TCP
SMTP	25	TCP
DNS	53	UDP (TCP for zone transfer)
HTTP	80	TCP
POP3	110	TCP
IMAP	143	TCP
HTTPS	443	TCP

A25. What are the three parts of a web browser? [Notes]

- **Controller** - receives input from the keyboard and mouse and uses the other two parts.
- **Client protocols** - the set of protocols such as HTTP and FTP that fetch the document.
- **Interpreters** - HTML, Java or JavaScript interpreters that display the document on screen.

A26. Why does DNS use UDP rather than TCP? [Notes]

- A name query and its answer are **one short request and one short reply** - a full TCP handshake would cost more than the query.
- UDP is **connectionless and fast**, and the resolver simply retries if no answer arrives.
- TCP is used only for large transfers such as zone replication between name servers.

A27. Explain the roles of the user agent and the message transfer agent in e-mail. [Notes]

- **User agent** - the mail program (Outlook, Thunderbird, webmail) that lets a person compose, read and manage mail; it hands outgoing mail to the queue.
- **Message transfer agent** - the server-side software that moves mail across the network using SMTP on port 25, and stores it in the recipient's mailbox.
- The **mail queue** holds outgoing mail while the MTA retries a temporary failure.

Section B | 7-mark questions

Write the heading points first, then expand each one. The bullet structure below is the marking scheme: one bullet is worth roughly half a mark to a mark.

B1. Discuss the client-server model architecture and state the functionality of FTP with a suitable diagram. [PYQ 2022; PYQ 2024-25]

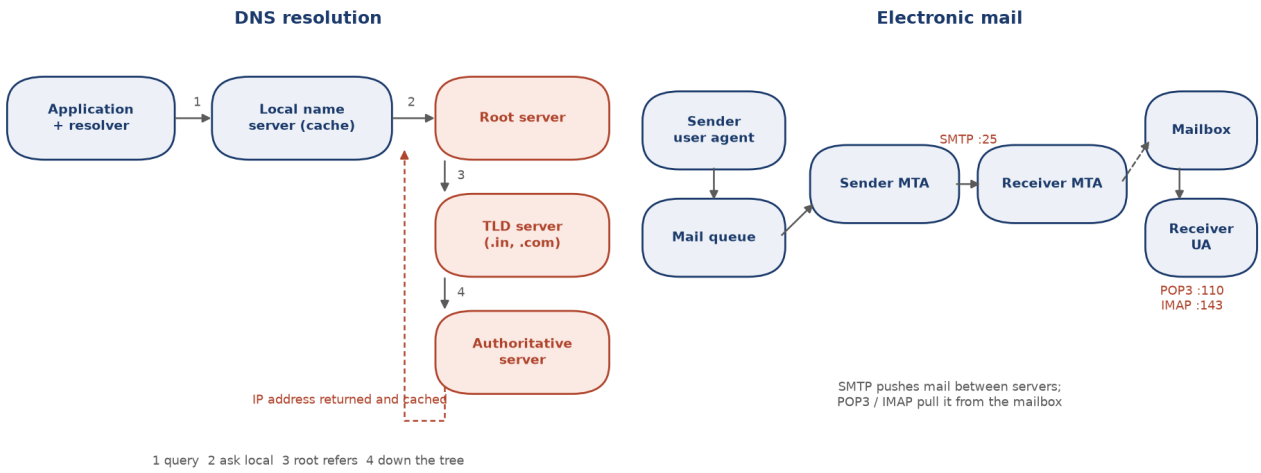
- **Client** - a program on the local machine that requests a service; **server** - a program on a remote machine that provides it. One server serves many clients at once.
- **Advantages**: centralised backup and administration, central security, better performance and easy scalability.
- **Disadvantages**: congestion at the server, a single point of failure, and data that may be modified in transit.
- **Socket interface**: a process reaches the network through a socket, identified by the pair (IP address, port).
- **FTP** is the classic client-server application: the client runs on the local host, the server on the remote host, and files are copied in either direction.
- **Two connections**: a control connection on TCP port 21 that stays open for the session and carries commands and replies, and a data connection on TCP port 20 opened for each file transfer.
- **Why two**: control and data travel independently, so a transfer can be aborted, and commands can be sent while data are moving; it also removes the need for a header on every data block.
- **What it hides**: differences in file conventions, text encoding and directory structure between the two systems.

Diagram to draw: two boxes - local host (user interface, client control process, client data transfer process) and remote host (server control process, server data transfer process); two labelled arrows for the control and data connections.

B2. Define DNS. List the types of name servers and explain the working of DNS. [PYQ 2022; PYQ 2023; PYQ 2024-25]

- **DNS** is a hierarchical, client-server naming system that maps domain names to IP addresses; it uses UDP port 53 and a domain-based hierarchical naming scheme.
- **Domain name space** is divided into generic domains (.com, .org, .edu), country domains (.in, .us, .uk) and the inverse domain (in-addr.arpa) for reverse lookup.
- **Name servers**: root server, primary (authoritative) server, secondary server and local (caching) server.

1. The application calls the **resolver** with a name query.
2. The resolver checks its cache; on a hit it returns the address immediately.
3. On a miss it queries the **local name server**, which checks its cache and zone file.
4. If that server cannot answer, it becomes a client: it asks the **root server**, which points to the top-level domain server, which points to the **authoritative server** for the domain.
5. The answer travels back along the same chain, and every server on the way **caches** it for its TTL.
6. The resolver returns the IP address to the application, which can now open a TCP connection.



Two query styles: recursive (the server does the work and returns the final answer) and iterative (the server returns a referral and the client asks the next server itself).

B3. Explain electronic mail: the architecture of SMTP with a diagram, and the protocols used to send and read mail. [PYQ 2022]

1. The **user agent** on the sender's machine composes the message and puts it in the **mail queue**.
2. The sender's **MTA** opens a **TCP connection to port 25** of the receiving MTA and pushes the message with SMTP commands.
3. The receiving MTA stores it in the recipient's **mailbox**.
4. The recipient's user agent retrieves it with a **mail access protocol** - POP3 on port 110, or IMAP on port 143.
5. MIME extends SMTP so that images, audio and attachments can be carried, since SMTP itself is a pure text protocol.

- **Push versus pull**: SMTP pushes mail from client to server and server to server; POP3 and IMAP pull it from the mailbox to the user.
- **Diagram to draw**: user at a terminal, user agent, queue, sender MTA, TCP port 25, receiver MTA, mailbox, receiver user agent, user at a terminal - with POP3 or IMAP on the last hop.

Ports to remember: SMTP 25 (push), POP3 110, IMAP 143 (pull). Secure versions: 587, 995 and 993.

B4. Give short notes on (a) HTTP and (b) SMTP, and explain the relationship of HTTP with the WWW. [PYQ 2023; PYQ 2022; PYQ 2024-25]

- **HTTP** - the protocol used to access data on the World Wide Web; it uses TCP on port 80 and behaves as a combination of FTP and SMTP.
- Two message types, request and response, each with a start line, headers and an optional body.
- It can be **non-persistent** (a new TCP connection per object, HTTP/1.0) or **persistent** (one connection for many objects, HTTP/1.1).
- **SMTP** - the protocol that sends e-mail between servers; TCP port 25, connection-oriented, pure text, extended by MIME.
- **WWW and HTTP**: the web is the set of linked hypertext documents; HTTP is how a browser fetches them. The URL gives the address, the browser is the HTTP client, the web server is the HTTP server, and the returned object is interpreted and displayed.
- **Proxy servers** sit between browser and server: they cache objects, filter access and hide the client's address.

Point	HTTP	SMTP
Purpose	Retrieve web objects	Transfer mail
Port	80	25
Direction	Pull - client asks	Push - client sends

Point	HTTP	SMTP
Content	Text, hypertext, audio, video	Text only, plus MIME

B5. Explain HTTP in detail: messages, connections and proxy servers. [PYQ 2022; PYQ 2023]

- **Request message** - request line with method, URL and version, then headers (Host, User-Agent, Accept, Cookie), then an optional body for POST.
- **Response message** - status line with version and status code, then headers (Server, Content-Type, Content-Length, Set-Cookie), then the object.
- **Methods**: GET (retrieve), POST (send data to the server), PUT (store), DELETE (remove), HEAD (headers only).
- **Status codes**: 1xx information, 2xx success, 3xx redirection, 4xx client error, 5xx server error - 200 and 404 are the two you must recognise.
- **Non-persistent** connection: one object per TCP connection, so a page with ten images costs eleven connections; **persistent** keeps one connection for all of them, which is the HTTP/1.1 default.
- **Proxy server**: an intermediary that caches responses, filters requests and hides the client; it acts as a server to the client and as a client to the origin server.
- **Cookies** let a stateless protocol keep state across requests.

B6. Explain FTP in detail: its two connections, commands and the problems it solves. [PYQ 2024-25; PYQ 2022]

1. **Control connection** - client opens TCP port 21 to the server and keeps it open for the whole session; it carries commands and replies in NVT ASCII.
2. **Data connection** - opened on TCP port 20 for each file transfer and closed when that file is complete.
3. **Commands**: RETR (retrieve), STOR (store), LIST (list directory), CWD (change directory), ABOR (abort), QUIT.
4. **Login**: USER and PASS, or anonymous FTP for public files.
5. **Three problems it solves**: different file naming conventions, different text and data representations, and different directory structures - FTP defines a common view for both ends.

- **Why the control connection stays open**: commands such as ABOR or STAT must be possible while data are moving, and replies must be unambiguous.
- **Active versus passive mode**: in active mode the server opens the data connection to the client, which firewalls block; in passive mode the client opens it, which is why passive mode is the default today.

B7. Explain remote login (Telnet) and the role of NVT and the pseudoterminal driver. [Notes]

1. Local login means logging into a local time-sharing system; **remote login** means using an application on a distant machine - Telnet is the general-purpose client-server program for it.
2. The local operating system accepts the keystrokes but does not interpret them; they go to the Telnet client.
3. The client converts them to **Network Virtual Terminal** characters - a universal, terminal-independent set - and hands them to the TCP/IP stack.
4. The characters cross the Internet and reach the remote stack, where the Telnet server converts them into the remote system's characters.
5. The remote operating system expects input from a **terminal driver**, so a **pseudoterminal driver** is inserted: it pretends the characters came from a terminal, and the operating system passes them to the application.

Why NVT: there are hundreds of terminal types. Translating everything into one intermediate set means only 2 conversions are needed instead of one per pair of terminal types.

B8. Explain the application-layer protocols of this unit with their ports in one table, and say which transport each uses. [Notes]

Protocol	Function	Port	Transport
DNS	Name to address resolution	53	UDP
SMTP	Send mail between servers	25	TCP
POP3	Download mail from the mailbox	110	TCP

Protocol	Function	Port	Transport
IMAP	Access and synchronise mail on the server	143	TCP
FTP	Transfer files	20 and 21	TCP
Telnet	Remote login	23	TCP
HTTP	Retrieve web objects	80	TCP
HTTPS	HTTP over TLS	443	TCP

- **Why TCP for most of them**: they carry data whose loss or reordering matters, so they want reliability and ordering.
- **Why UDP for DNS**: one short query and reply, where the cost of a handshake exceeds the cost of the query itself.